LOOPS IN C++

NESTED FOR LOOPS



LECTURE OUTLINE

Nested loop

Break

Continue

NESTED LOOPS

- *A loop can be nested inside another loop.*
- *Nested loops* consist of an outer loop and one or more inner loops
- Each time the outer loop is repeated, the inner loops are reentered, and started anew.

CODE

```
#include<iostream>
using namespace std;
main()
{
    for(int num = 1; num <= 5; num++)
    {
        for(int t = 1; t <= 5;t++)
        {
            cout<<num<<" ";
        }
        cout<<endl;
    }
}
```

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

ACTIVITY

Show the output of the following programs. (*Tip*: Draw a table and list the variables in the columns to trace these programs.)

```
for (int i = 1; i < 5; i++)
{
  int j = 0;
  while (j < i)
  {
    cout << j << " ";
    j++;
  }
}
```

0
0 1
0 1 2
0 1 2 3

(a)

```
int i = 0;
while (i < 5)
{
  for (int j = i; j > 1; j--)
    cout << j << " ";
  cout << "*****" << endl;
  i++;
}
```

*****
*****
2 *****
3 2 *****
4 3 2 *****

(b)

ACTIVITY

```
int i = 5;
while (i >= 1)
{
  int num = 1;
  for (int j = 1; j <= i; j++)
  {
     cout << num << "xxx";
     num *= 2;
  }

  cout << endl;
  i--;
}
```

```
*****
*****
2 *****
3 2 *****
4 3 2 *****
```

(c)

```
int i = 1;
do
{
  int num = 1
  for (int j = 1; j <= i; j++)
  {
     cout << num << "G";
     num += 2;
  }

  cout << endl;
  i++;
} while (i <= 5);
```

(d)

BREAK AND CONTINUE

- The break and continue keywords provide additional controls in a loop.
- You have used the keyword break in a switch statement.
- You can also use break in a loop to immediately terminate the loop.

```
#include <iostream>
using namespace std;

int main()
{
  int sum = 0;
  int number = 0;

  while (number < 20)
  {
    number++;
    sum += number;
    if (sum >= 100)
      break;
  }

  cout << "The number is " << number << endl;
  cout << "The sum is " << sum << endl;

  return 0;
}
```

```
#include <iostream>
using namespace std;

int main()
{
  int sum = 0;
  int number = 0;

  while (number < 20)
  {
    number++;
    if (number == 10 || number == 11)
      continue;
    sum += number;
  }

  cout << "The sum is " << sum << endl;

  return 0;
}
```

USING BREAK IN A NESTED LOOP

- In a nested loop, the break statement only interrupts the loop it is placed in.

```
for (int row = 0; row < 5; row++)
{
     for (int star = 0; star < 20; star++)
     {
          cout << '*';
          if (star == 10)
               break;
     }
     cout << endl;
}
```

The output of the program segment above is:

```
**********
**********
**********
**********
**********
```

ACTIVITY

Write a C++ program that takes an integer **N** as input and prints a pyramid pattern of asterisks (*) with **N** rows. The pyramid should be center-aligned, with each row containing an increasing number of asterisks in the shape of a triangle.
Output:

```
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

ACTIVITY

```
    for (int i = 1; i <= n; i++) {
        // Print spaces
        for (int j = 1; j <= n - i; j++) {
            cout << "  ";
        }
        // Print asterisks
        for (int k = 1; k <= (2 * i - 1); k++) {
            cout << " * ";
        }
        cout << endl;
    }

    return 0;
}
```

NEXT LECTURE

Arrays in C++